CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

# 7.4 Electromagnetic Spectrum

**Syllabus Section:** 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

## Question 1 Easy [1 mark]

Which of the following is the approximate range of frequencies of infrared radiation?

**A** $1 \times 10^{14} – 1 \times 10^{16}$ Hz

**B** $1 \times 10^{12} – 1 \times 10^{14}$ Hz

**C** $1 \times 10^{9} – 1 \times 10^{11}$ Hz

**D** $1 \times 10^{3} – 1 \times 10^{9}$ Hz

### Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The range of **infrared** radiation is $3.0 \times 10^{11} – 4.3 \times 10^{14}$ Hz

**A** is incorrect because the **ultraviolet** light range is $7.5 \times 10^{14} – 3.0 \times 10^{16}$ Hz

**C** is incorrect because the **microwave** range is $3.0 \times 10^{9} – 3.0 \times 10^{11}$ Hz

**D** is incorrect because the **radio** range is less than $3.0 \times 10^{9}$ Hz

**Question 2** Easy [1 mark]

Which of the following statements is a correct description of part of the electromagnetic spectrum?

A waves of wavelength $5 \times 10^{-7}$ m are infrared waves

B waves of wavelength $5 \times 10^{-9}$ m are high energy gamma rays

C waves of wavelength $9 \times 10^{-7}$ m are light waves

D waves of wavelength $3 \times 10^{-8}$ m are ultraviolet waves

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- Waves with a wavelength in the range of $4 \times 10^{-7}$ to $10^{-8}$ m are **ultraviolet**

**A** is incorrect because waves with a wavelength in the range $4 - 7 \times 10^{-7}$ m are visible light

**B** is incorrect because waves with a wavelength in the range $10^{-8} - 4 \times 10^{-13}$ m are x-rays

**C** is incorrect because waves with a wavelength in the range $10^{-3} - 7 \times 10^{-7}$ are infra-red

**Question 3** Easy [1 mark]

Which of the following give the order of magnitude of frequencies of the visible part of the electromagnetic spectrum?

A $10^{15}$ Hz

B $10^{14}$ Hz

C $10^{13}$ Hz

D $10^{16}$ Hz

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- $10^{14}$ Hz is in the visible light range of 4.3 – 7.5 × $10^{14}$ Hz

**A** is incorrect because $10^{15}$ Hz is in the **ultraviolet** light range: 7.5 × $10^{14}$ – 3.0 × $10^{16}$ Hz

**C** is incorrect because $10^{13}$ Hz is in the **infrared** range: 3.0 × $10^{11}$ – 4.3 × $10^{14}$ Hz

**D** is incorrect because $10^{16}$ Hz is in the **x-ray** range: 3.0 × $10^{16}$ – 7.5 × $10^{20}$ Hz

**Question 4** Easy [1 mark]

The diagram shows part of the electromagnetic spectrum.




Which row correctly labels the regions marked P and Q?

| | P | Q |
|---|---|---|
| **A** | infrared | x-ray |
| **B** | ultraviolet | x-ray |
| **C** | x-ray | infrared |
| **D** | infrared | microwaves |

**A**

**B**

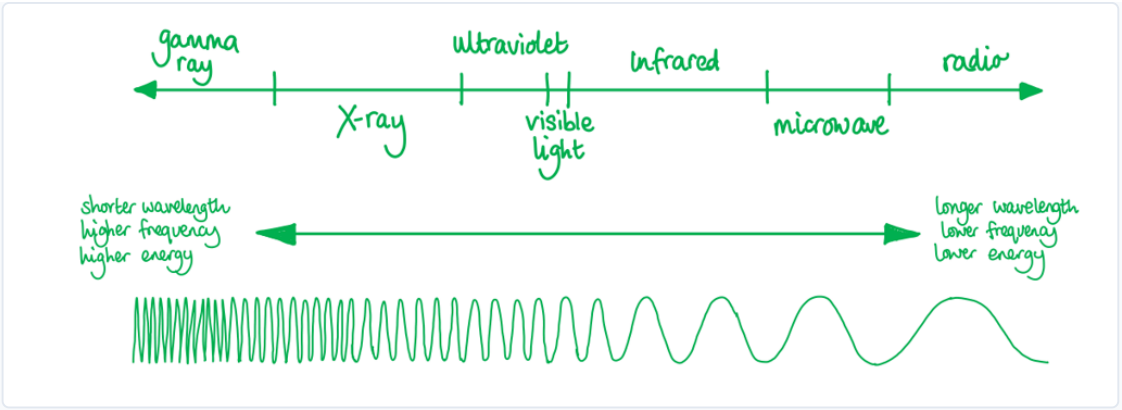
**C**

**D**

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The electromagnetic spectrum is a term used to describe waves created as a result of vibrations between an electric field and a magnetic field
- The spectrum is made up of:




- P is on the high-frequency, short-wavelength side = X-ray
- Q is on low-frequency side = infrared

*Make sure to check the labelling for which end is high frequency and low frequency.*

**Question 5** Easy [1 mark]

Which of the following do **not** travel at the speed of light in a vacuum?

A x-rays

B radio waves

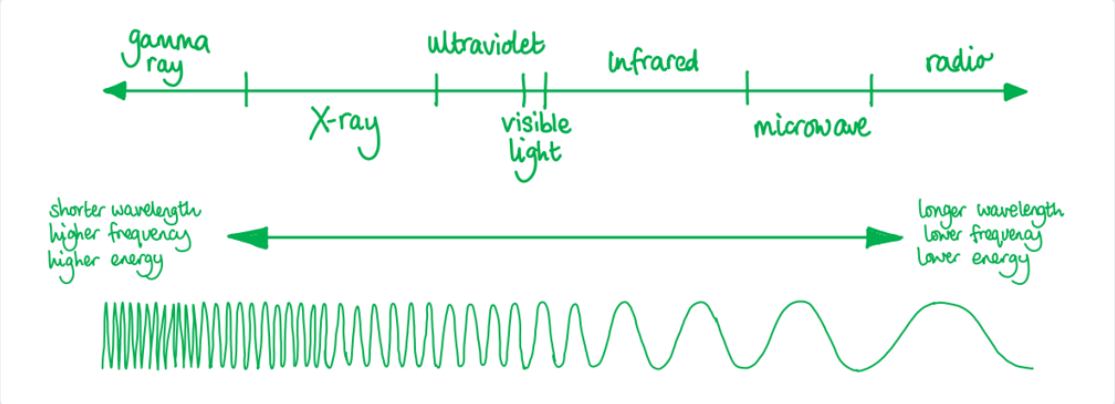
C microwaves

D electrons

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The electromagnetic spectrum is a term used to describe waves created as a result of vibrations between an electric field and a magnetic field.
- The electromagnetic spectrum is made up of




- The speed of all EM waves in a vacuum is $3.0 \times 10^8$ m $s^{-1}$
- Electrons do not form part of the EM spectrum, and the speed that they travel in a vacuum is roughly one one-hundredth the speed of light

All the other options are part of the electromagnetic spectrum so all travel at the same speed in a vacuum

**Question 6** Easy [1 mark]

Which of the following is the number of wavelengths in one metre of visible light?

A $10^4$

B $10^6$

C $10^8$

D $10^{10}$

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- The range of wavelength of visible light is typically 400 – 700 nm
- If we take the average value of this range: 550 nm = 550 × $10^{-9}$ m
- The number of wavelengths in one metre is equal to
  - $\frac{1}{550 \times 10^{-9}} = 1.8 \times 10^6$
- Therefore, there are around $10^6$ wavelengths of visible light in 1 m

**A** is incorrect because this corresponds to a wavelength of $\frac{1}{10^4} = 10^{-4}$ which corresponds to **infra-red** light

**C** is incorrect because this corresponds to a wavelength of $\frac{1}{10^8} = 10^{-8}$ which corresponds to **ultraviolet** light

**D** is incorrect because this corresponds to a wavelength of $\frac{1}{10^{10}} = 10^{-10}$ which corresponds to **x-rays**

**Question 7** Medium [1 mark]

Two lasers are set up to pass through a vacuum. One laser emits red light; the other emits green light.

Which property of the two lasers are different?

**A** speed

**B** plane of polarisation

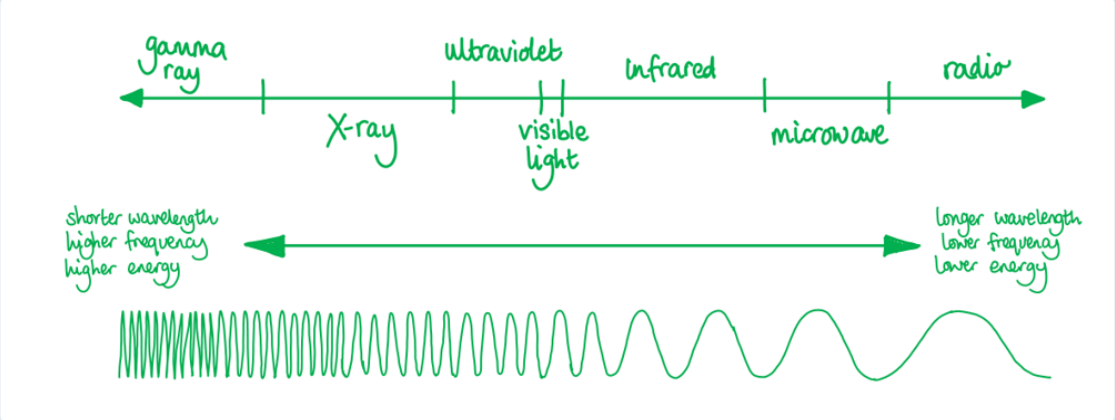
**C** frequency

**D** amplitude

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Visible light is part of the electromagnetic spectrum




- A laser is a highly directional beam of light
- The **frequency** of the light determines the colour of the laser light
- Since red and green are on different parts of the spectrum, they have different frequencies

**A** is incorrect because the speed of all EM waves in a vacuum is the same ($3.0 \times 10^8$ m $s^{-1}$)

**B** is incorrect because laser light is not polarised light but light directed into a narrow beam of light, polarised light is light that is travelling in one plane only

**D** is incorrect because the amplitude of the light wave determines the intensity or brightness of the wave, it tells you how much energy the wave carries

**Question 8** Medium [1 mark]

The wavelength of light emitted by a laser is 600 nm.

Which of the following is the distance in number of wavelengths, that is travelled by light in one second?

**A** $5.0 \times 10^{14}\ s^{-1}$

**B** $5.0 \times 10^{8}\ s^{-1}$

**C** $5.0 \times 10^{11}\ s^{-1}$

**D** $5.0 \times 10^{17}\ s^{-1}$

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- Wavelength is defined as the distance between successive crests of a wave
- Frequency of a wave is defined as the number of waves passing a point in one second (the same as the definition given in the question)
- The wave equation that links speed, wavelength and frequency is:
  - $v = \lambda \times f$
- The speed of light is $3.0 \times 10^{8}$ m $s^{-1}$
- The wavelength is $6.0 \times 10^{-7}$ m
  - Therefore, the frequency is equal to $= \frac{3.0 \times 10^{8}}{6.0 \times 10^{-7}} = 5.0 \times 10^{14}$ Hz

**Question 9** Medium [1 mark]

Waves of the electromagnetic spectrum travel through a vacuum at speed $c$ and with a wavelength $\lambda$ and frequency $f$

Which of the following describes the speed and wavelength when the frequency is $\frac{f}{2}$ ?

| | $\lambda$ | $c$ |
|---|---|---|
| **A** | $2\lambda$ | $2c$ |
| **B** | $2\lambda$ | $c$ |
| **C** | $\frac{\lambda}{2}$ | $c$ |
| **D** | $\frac{\lambda}{2}$ | $\frac{c}{2}$ |

A

B

C

D

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Wavelength is defined as the distance between successive crests of a wave
- Frequency of a wave is defined as the number of waves passing a point in one second
- The wave equation that links speed, wavelength and frequency is:
  - $v = \lambda \times f$
- The speed $c$ of all EM waves in a vacuum is constant
- So, a change in $f$ will only cause $\lambda$ to change
  - $c = f\lambda$
  - $c = \frac{f}{2}2\lambda$
- Therefore, if the frequency of the wave is halved, then the wavelength will double

**Question 10** Medium [1 mark]

The diameter of a nucleus and the wavelength of an electromagnetic wave are in the same order of magnitude.

In which region of the electromagnetic spectrum would that wavelength occur?

**A** infrared

**B** visible light

**C** x-ray

**D** gamma-ray

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The diameter of a hydrogen nucleus is $1.76 \times 10^{-15}$m
- The wavelength of a gamma-ray is in the range of $10^{-10}$ to $10^{-16}$ m
- Therefore, gamma rays have a wavelength similar to the diameter of a hydrogen nucleus

**A** is incorrect because **infrared** waves have a wavelength in the range: $10^{-3}$ to $7 \times 10^{-7}$ m

**B** is incorrect because **visible light** has waves with a wavelength in the range: $4 \times 10^{-7}$ to $7 \times 10^{-7}$ m

**C** is incorrect because **x-rays** have a wavelength in the range: $10^{-8}$ to $4 \times 10^{-13}$ m

## Question 11 Medium [1 mark]

R is a source emitting infra-red radiation and S is a source emitting ultra-violet radiation. The table suggests the wavelengths for R and S.

Which row is correct?

| | Wavelength emitted by R / m | Wavelength emitted by S / m |
|---|---|---|
| **A** | $5 \times 10^{-7}$ | $5 \times 10^{-8}$ |
| **B** | $5 \times 10^{-8}$ | $5 \times 10^{-10}$ |
| **C** | $5 \times 10^{-5}$ | $5 \times 10^{-8}$ |
| **D** | $5 \times 10^{-5}$ | $5 \times 10^{-10}$ |

A

B

C

D

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Source R is emitting infra-red radiation; infra-red radiation has a wavelength in the range of $10^{-3}$ to $7 \times 10^{-7}$ m
- Source S is emitting ultra-violet radiation, ultra-violet radiation has a wavelength in the range of $4 \times 10^{-7}$ to $10^{-8}$ m
- The only answer that would fit both of these ranges is option **C**

**A** is incorrect because R is in the wavelength range of visible light and S is in the wavelength range of ultra-violet radiation

**B** is incorrect because R is in the range of ultra-violet radiation and S in the range of the wavelengths of x-rays

**D** is incorrect because R is in the correct range, but S is in the range of the wavelengths of x-rays

**Question 12** Medium [1 mark]

Which statement would be deduced from a table of wavelengths of the waves in the electromagnetic spectrum?

A the wavelength range for radio waves is less than that for infrared waves

B the wavelength range for x-rays is less than that for radio waves

C red light has a shorter wavelength than green light

D green light has a shorter wavelength than x-rays

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Wavelength is defined as the distance between successive crests of a wave
  - The wavelength of x-rays is in the range $1 \times 10^{-8}$– $1 \times 10^{-13}$ m
  - The wavelength of radio waves is greater than $1 \times 10^{-1}$ m
- Therefore, the wavelength range for x-rays is less than that for radio waves

**A** is incorrect because the wavelength of radio waves is greater than $1 \times 10^{-1}$ m, the wavelengths of infrared waves is in the range of $1 \times 10^{-3} - 7 \times 10^{-7}$ m

**C** is incorrect because the wavelength of red light is 620 – 750 nm whereas green light is shorter with a range of 495 – 570 nm

**D** is incorrect because the wavelength of green light is in the range of 495 – 570 nm, whereas x-rays are in the range of $1 \times 10^{-8} - 4 \times 10^{-13}$ m

The key to answering this question correctly is recognising that options A and B use the word **range**.

**Question 13** Medium [1 mark]

A telescope detects electromagnetic radiation with wavelengths 2 cm.

Which type of telescope is it ?

**A** x-ray telescope

**B** optical telescope

**C** microwave telescope

**D** radio telescope

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Wavelength is defined as the distance between successive crests of a wave
  - Wavelengths of 2 cm are 0.02 m or $2 \times 10^{-2}$ m
  - The wavelengths of microwaves are in the range of $0.1 – 1 \times 10^{-3}$ m
- Therefore, the telescope detects microwave radiation

**A** is incorrect because x-rays are in the range: $1 \times 10^{-8} – 4 \times 10^{-13}$ m

**B** is incorrect because visible light rays are in the range: $4 \times 10^{-7} – 7 \times 10^{-7}$ m

**D** is incorrect because radio waves have wavelengths greater than $1 \times 10^{-1}$ m

Most homes use a microwave oven. Consider the possible size of the microwaves used inside the oven.

**Question 14** Medium [1 mark]

P is a source emitting radio waves and Q is a source emitting X-rays. The table suggests the wavelengths for P and Q.

Which row is correct?

| | **Wavelength emitted by P / m** | **Wavelength emitted by Q / m** |
|---|---|---|
| **A** | $3 \times 10^{-1}$ | $3 \times 10^{-11}$ |
| **B** | $3 \times 10^{-7}$ | $3 \times 10^{-15}$ |
| **C** | $3 \times 10^{-2}$ | $3 \times 10^{-15}$ |
| **D** | $3 \times 10^{-7}$ | $3 \times 10^{-11}$ |

**A**

**B**

**C**

**D**

**Mark Scheme & Explanation:**

*The correct answer is A*

- Source P is emitting radio waves
  - Radio waves span a very wide range, typically from about $10^{-1}$ to $10^{3}$ m and beyond
  - This, $3 \times 10^{-1}$ m is in the radio wave region
- Source Q is emitting X-rays
  - X-rays have a wavelength in the range of $10^{-8}$ to $10^{-13}$ m
- The only answer that would fit both of these ranges is option **A**

**B** is incorrect because P is in the wavelength range of UV and Q is in the wavelength range of gamma waves

**C** is incorrect because P is in the correct range, but Q is in the range of the wavelengths of gamma waves

**D** is incorrect because P is in the range of UV and Q is in the correct range

**Question 15** Hard [1 mark]

An infrared wave has a wavelength of 30 μm and frequency $f_{\text{infrared}}$. An x-ray has a wavelength of 400 pm and frequency $f_{\text{x-ray}}$.

What is the value of the ratio $\frac{f_{x\text{-}ray}}{f_{infrared}}$?

**A** $1.3 \times 10^{-5}$

**B** 0.075

**C** 13

**D** $7.5 \times 10^{4}$

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The known quantities are
  - Wavelength of infrared wave, $\lambda = 30 \times 10^{-6}$ m
  - Wavelength of x-ray, $\lambda = 400 \times 10^{-12}$ m
- The wave equation is
  - $c = f\lambda$
  - $f = \frac{c}{\lambda}$
- The frequencies of the infrared wave and x-rays are
  - $f_{infrared} = \frac{c}{3.0 \times 10^{-6}}$
  - $f_{x\text{-}ray} = \frac{c}{400 \times 10^{-12}}$
- The value of the ratio $\frac{f_{x\text{-}ray}}{f_{infrared}}$ is
  - $\frac{f_{x\text{-}ray}}{f_{infrared}} = \frac{c}{400 \times 10^{-12}} \times \frac{30 \times 10^{-6}}{c}$
  - $\frac{f_{x\text{-}ray}}{f_{infrared}} = \frac{30 \times 10^{-6}}{400 \times 10^{-12}} = 7.5 \times 10^{4}$
- This is option **D**

**Question 16** Hard [1 mark]

Fig. 1.1 shows the percentage of transmitted wavelengths which can travel through Earth's atmosphere and reach sea level.

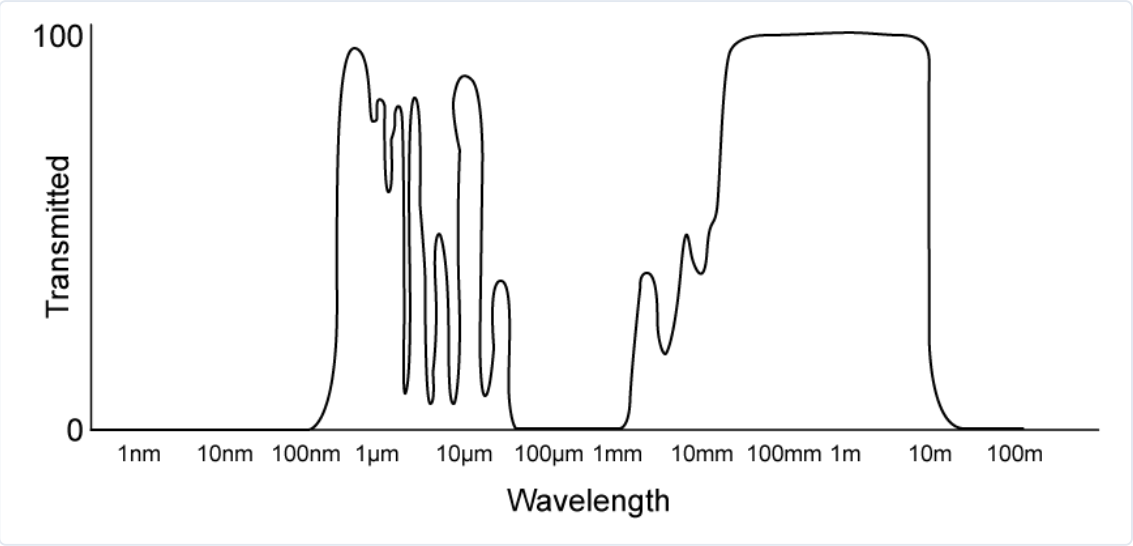



**Fig. 1.1**

What is the range of frequencies that can penetrate Earth's atmosphere and reach sea level?

**A** $10 \text{ nHz} \leq f \leq 100 \text{ μHz}, \quad 1 \text{ mHz} \leq f \leq 10 \text{ Hz}$

**B** $10 \text{ Hz} \leq f \leq 100 \text{ Hz}$

**C** $30 \text{ MHz} \leq f \leq 300 \text{ GHz}, \quad 3 \text{ THz} \leq f \leq 3000 \text{ THz}$

**D** $30 \text{ MHz} \leq f \leq 300 \text{ THz}$

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The wave equation is
  - $c = f\lambda$
  - $f = \frac{c}{\lambda}$
- The wavelengths that are transmitted through Earth's atmosphere and reach sea level are those which have a transmitted % above zero
  - This is wavelengths 100 nm to 100 μm and 1 mm to 10 m

- When $\lambda$ = 100 nm = $100 \times 10^{-9}$ m
  - $f = \frac{3 \times 10^8}{100 \times 10^{-9}}$
  - $f = 3 \times 10^{15}$ = 3000 THz
- When $\lambda$ = 100 μm = $100 \times 10^{-6}$ m
  - $f = \frac{3 \times 10^8}{100 \times 10^{-6}}$
  - $f = 3 \times 10^{12}$ = 3 THz
- When $\lambda$ = 1 mm = $1 \times 10^{-3}$ m
  - $f = \frac{3 \times 10^8}{1 \times 10^{-3}}$
  - $f = 3 \times 10^{11}$ = 300 GHz
- When $\lambda$ = 10 m
  - $f = \frac{3 \times 10^8}{10}$
  - $f = 3 \times 10^7$ = 30 MHz
- Therefore the range of electromagnetic frequencies that can penetrate Earth's atmosphere is
  - $30 \ MHz \leq f \leq 300 \ GHz, \ 3 \ THz \leq f \leq 3000 \ THz$
- This is option **C**
- Although this seems like a lot of calculations, you can figure out the correct answer by process of elimination
- Between 100 μm and 1 mm there are no electromagnetic waves reaching sea level
  - Therefore there must be two sets of ranges
  - This rules out option **B** and **D**
- Option **A** must be incorrect as the values of the frequencies match the wavelengths from the graph
  - As $f = \frac{c}{\lambda}$ the values from the graph cannot match the values of the frequencies

**Question 17** Hard [1 mark]

Monochromatic light is incident normally upon a double slit where the distance between the slits is 0.02 mm. The angle between the second-order and central maxima is 3.96°.

What is the colour of the monochromatic light?

A Violet

B Blue

C Green

D Red

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The known quantities are
  - Distance between the slits, $d = 0.02 \times 10^{-3}$ m
  - Angle between the first-order and central maxima, $\theta = 3.96°$
- The diffraction grating equation is given by
  - $d \sin\theta = n\lambda$
- When $n = 2$ for the second-order maxima
  - $2\lambda = d \sin\theta$
  - $\lambda = \frac{d \sin\theta}{2}$
  - $\lambda = \frac{0.02 \times 10^{-3} \times \sin 3.96}{2} = 690$ nm (2 s.f)
- The range of wavelengths for the colour red is around 620 nm to 700 nm
  - Therefore, the colour of the monochromatic light is red
- This is option **D**
- You are not expected to know the wavelength ranges for the individual colours for visible light
- For this particular question you can use a process of elimination to determine the correct answer
  - The range of wavelengths for the visible light spectrum is between 400 nm and 700 nm
  - Violet and blue have the shortest wavelengths and red has the largest wavelength
  - Green generally sits in the middle of the visible light wavelength range

- 690 nm is close to the largest wavelength of visible light, so the answer is red

**Question 18** Hard [1 mark]

The table below contains the frequencies of various parts of the electromagnetic spectrum.

Which row correctly describes X as infrared and Y as X-rays?

| | X | Y |
|---|---|---|
| **A.** | $3 \times 10^{5}$ Hz | $3 \times 10^{20}$ Hz |
| **B.** | $3 \times 10^{10}$ Hz | $3 \times 10^{16}$ Hz |
| **C.** | $3 \times 10^{18}$ Hz | $3 \times 10^{14}$ Hz |
| **D.** | $3 \times 10^{14}$ Hz | $3 \times 10^{19}$ Hz |

A

B

C

D

**Mark Scheme & Explanation:**

*The correct answer is **D** because:*

- The speed, frequency and wavelength of an electromagnetic wave is defined by:
  - $c = f\lambda$
- Rearranging for $\lambda$ gives:
  - $\lambda = \frac{c}{f}$
  - $c = 3.00 \times 10^{8}$ m s$^{-1}$
- X = $3 \times 10^{14}$ Hz
  - $\lambda = \frac{3.00 \times 10^{8}}{3 \times 10^{14}}$
  - Therefore, $\lambda = 1 \times 10^{-6}$ m
- Infrared wavelengths vary between $1 \times 10^{-3}$ m and $7 \times 10^{-7}$ m
  - Therefore, X is infrared
- Y = $3 \times 10^{19}$ Hz

- $\lambda = \frac{3.00 \times 10^{8}}{3 \times 10^{19}}$
- Therefore, $\lambda = 1 \times 10^{-11}$ m

- X-ray wavelengths vary between $1 \times 10^{-8}$ m and $4 \times 10^{-13}$ m
  - Therefore, Y is an x-ray
- Answer **D** is the only answer that fits these criteria

**A** is incorrect as:

- X = $3 \times 10^{5}$ Hz, therefore $\lambda = 1 \times 10^{3}$ m which is a radio wave
- Y = $3 \times 10^{20}$ Hz, therefore $\lambda = 1 \times 10^{-12}$ m which is an x-ray

**B** is incorrect as:

- X = $3 \times 10^{10}$ Hz, therefore $\lambda = 1 \times 10^{-2}$ m which is a microwaves
- Y = $3 \times 10^{16}$ Hz, therefore $\lambda = 1 \times 10^{-8}$ m which is a UV wave

**C** is incorrect as:

- X = $3 \times 10^{18}$ Hz, therefore $\lambda = 1 \times 10^{-10}$ m which is an x-ray
- Y = $3 \times 10^{14}$ Hz, therefore $\lambda = 1 \times 10^{-6}$ m which is an infrared

If you need a reminder of the wavelengths of the EM spectrum:

| Electromagnetic Wave | Wavelength, λ (m) |
|---|---|
| Radio Waves | > 0.1 |
| Microwaves | $0.1 – 1 \times 10^{-3}$ |
| Infrared | $1 \times 10^{-3} – 7 \times 10^{-7}$ |
| Visible Light | $4 \times 10^{-7} – 7 \times 10^{-7}$ |
| Ultraviolet | $4 \times 10^{-7} – 1 \times 10^{-8}$ |
| X-rays | $1 \times 10^{-8} – 4 \times 10^{-13}$ |
| Gamma Rays | $4 \times 10^{-13} – 10^{-16}$ |